PlugPlan SG: Test Cases and Testing Results

SC2006 · Lab 4 · Author: Team SCE3-04 · Prepared with Claude Code · 10 November 2026

1. Summary

Suite	Technique	Target	Cases	Executed	Passed	Failed
tests/lab4.blackbox.test.ts	Equivalence class + boundary value	PlanEngine.calculatePlan	69	69	69	0
tests/lab4.basis-path.test.ts	Basis path, V(G) = 27	PlanEngine.failureForStation	27	26	26	0
tests/lab4.basis-path.test.ts	Basis path, V(G) = 18	PlanEngine.calculatePlan	18	17	17	0
Total			114	112	112	0

Automated with Vitest 4.1.11 on Node.js 22.12.0, Windows 11. Run from **PlugPlan SG/** with **npx vitest run tests/lab4**. Expected outputs were calculated by hand before execution; actual outputs were recorded by the test run. These suites alone cover 231/262 (88.16%) of engine.ts branches. The complete unit suite, including the 43 existing tests, passed 155/155. Two basis paths are infeasible and are explained below. No defect was found.

Traceability: FR and NFR numbers refer to the SRS. Requirements FR-05 to FR-19 realise use case UC-03 Create Charging Plan; the design class is PlanEngine; the implementation is src/lib/domain/engine.ts.

2. Black-box testing: PlanEngine.calculatePlan

PlanEngine is the control class that decides which charging stations are feasible and how they rank. Each row below is one input partition. Invalid classes are tested one at a time so a rejection names one field. Only boundaries and one representative per class are used, which keeps the suite small.

Input	Valid classes	Invalid classes	Boundary values	Cases
currentSoc (%)	[0, 100]	< 0; > 100	-0.1, 0, 100, 100.1	BB-01 to 05
Arrival battery energy	>= 0 kWh	< 0 kWh	0; -0.06	BB-06, 07
maxDetourMinutes; detour	[0, 120]; detour <= limit	< 0; > 120; detour > limit	-0.1, 0, 120, 120.1; limit 10 and 9.9	BB-08 to 13
maxChargingMinutes; time needed	[5, 720]; time <= limit	< 5; > 720; time > limit	4.9, 5, 720, 720.1; limit 8 and 7.9	BB-14 to 19
manualTargetSoc (%)	null; [automatic, 100]	< 0; > 100; below automatic	-0.1, 0, 53.3, 53.34, 80, 80.1, 100, 100.1	BB-20 to 28
reserveSoc (%)	[5, 40]	< 5; > 40	4.9, 5, 40, 40.1	BB-29 to 32
capacityKWh	[10, 250]	< 10; > 250	9.9, 10, 250, 250.1	BB-33 to 36
Destination + reserve energy	<= capacity	> capacity	60 kWh; 60.2 kWh	BB-37, 38
Direct trip energy left	>= reserve	< reserve	12 kWh; 11.9 kWh	BB-39, 40
Plug type	Match, including aliases	No match	n/a (categorical)	BB-41, 42
Opening hours	Window covers charging; overnight	Opens late; closes early	20:30-20:38; 20:31; 20:37	BB-43 to 46
Effective power (kW)	Connector or vehicle limit; >= minimum	< minimum	50; 50.1	BB-47 to 50
Price	per kWh, per hour, zero	negative; unsupported type	0; -0.01	BB-51 to 55
Strategy	Five presets	Any other value	n/a (categorical)	BB-56 to 61
Shortlist size	<= 3	> 3 is truncated	3; 4 stations	BB-62, 63
departureAt	ISO with or without offset	Impossible date	n/a	BB-64, 65
Snapshot age	< 10:00 fresh	>= 10:00 stale	9:59; 10:00	BB-66, 67
Fault expiry	> now is active	<= now is ignored	12:00:01; 12:00:00	BB-68, 69

Baseline, by hand: 60 kWh battery, 20 kWh/100 km, SOC 50%, reserve 20%, CCS2 DC connector 100 kW at S$0.30/kWh, vehicle DC limit 50 kW; 20 km/30 min to the station, 100 km/60 min onward, 200 km/90 min direct. Arrival 26 kWh (43.33%), target 32 kWh (53.33%), 6 kWh into the battery, 6.67 kWh from the charger, 8 min, S$2.00. The ranking fixture is four stations A to D; their totals are 108, 130, 126, 140 min, costs S$4.00, 2.67, 0.67, 3.33, availability 50%, 50%, 50%, 100%, and detours 10, 0, 20, 30 min. REJECTED names the field refused by input validation.

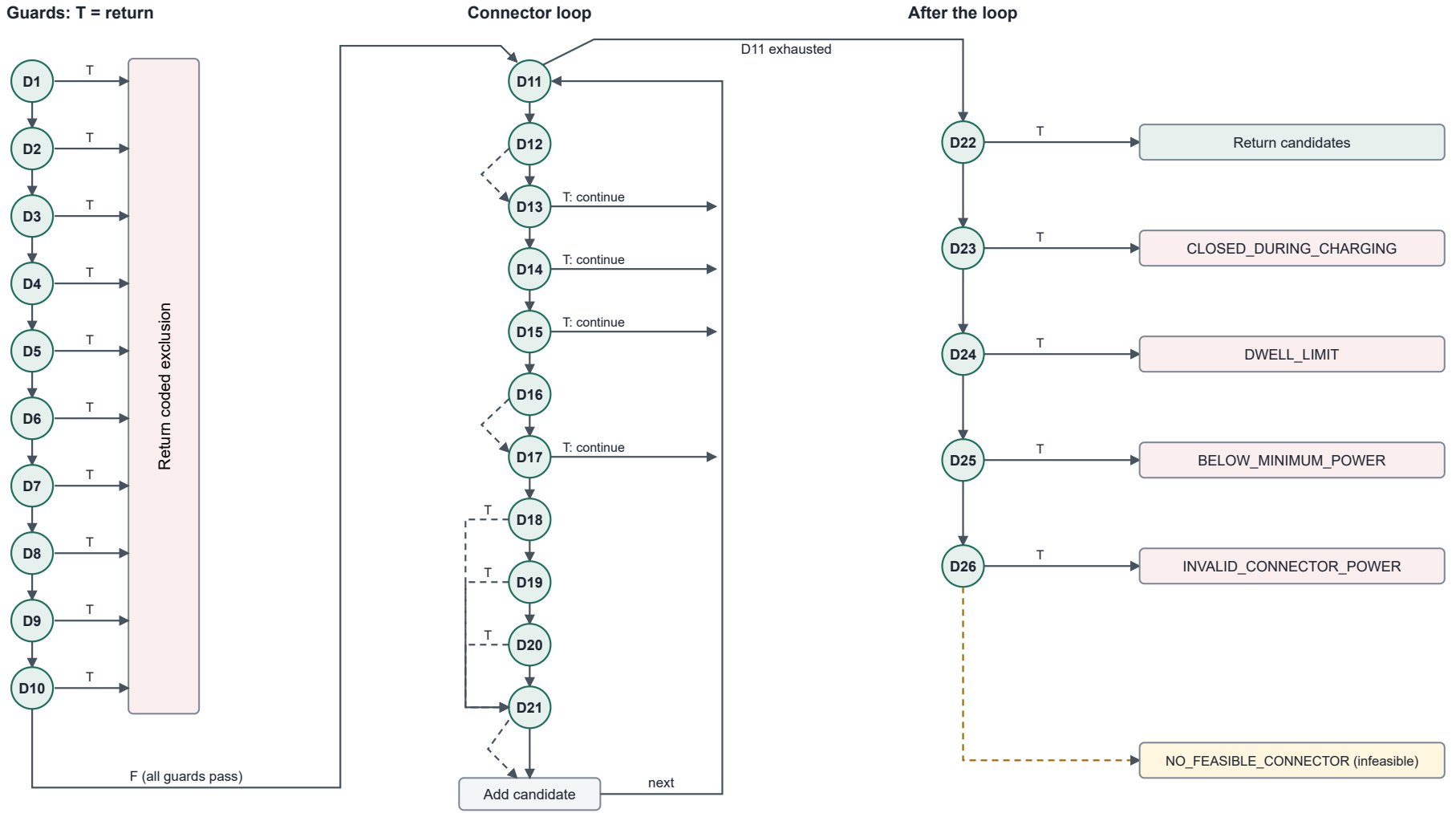
ID	Class / path	Trace	Test input	Expected output	Actual output	Result
BB-01	SOC invalid low (boundary-)	FR-05	currentSoc=-0.1	`REJECTED currentSoc`	`REJECTED currentSoc`	**PASS**
BB-02	SOC valid (lower boundary)	FR-05, FR-08	currentSoc=0; to-station 0 km	`arrivalSoc=0.00 targetSoc=53.33 batteryKWh=32.00 min=42.67 cost=10.67`	`arrivalSoc=0.00 targetSoc=53.33 batteryKWh=32.00 min=42.67 cost=10.67`	**PASS**
BB-03	SOC valid (nominal, numeric oracle)	FR-08, FR-13, FR-15, FR-17, NFR-04	Baseline: currentSoc=50	`arrivalSoc=43.33 targetSoc=53.33 batteryKWh=6.00 min=8.00 cost=2.00`	`arrivalSoc=43.33 targetSoc=53.33 batteryKWh=6.00 min=8.00 cost=2.00`	**PASS**
BB-04	SOC valid (upper boundary)	FR-05, FR-13	currentSoc=100	`EXCLUDED NO_CHARGING_REQUIRED`	`EXCLUDED NO_CHARGING_REQUIRED`	**PASS**
BB-05	SOC invalid high (boundary+)	FR-05	currentSoc=100.1	`REJECTED currentSoc`	`REJECTED currentSoc`	**PASS**
BB-06	Arrival energy = 0 (boundary, reachable)	FR-09	to-station 150 km; currentSoc=50	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-07	Arrival energy < 0 (boundary-)	FR-09	to-station 150 km; currentSoc=49.9	`EXCLUDED UNREACHABLE`	`EXCLUDED UNREACHABLE`	**PASS**
BB-08	Detour limit invalid low	FR-05	maxDetourMinutes=-0.1	`REJECTED maxDetourMinutes`	`REJECTED maxDetourMinutes`	**PASS**
BB-09	Detour limit lower boundary	FR-12	maxDetourMinutes=0; detour 0	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-10	Detour equals limit (boundary)	FR-12	maxDetourMinutes=10; detour 10	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-11	Detour above limit (boundary+)	FR-12	maxDetourMinutes=9.9; detour 10	`EXCLUDED DETOUR_LIMIT`	`EXCLUDED DETOUR_LIMIT`	**PASS**
BB-12	Detour limit upper boundary	FR-05	maxDetourMinutes=120	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-13	Detour limit invalid high	FR-05	maxDetourMinutes=120.1	`REJECTED maxDetourMinutes`	`REJECTED maxDetourMinutes`	**PASS**
BB-14	Charging limit invalid low	FR-05	maxChargingMinutes=4.9	`REJECTED maxChargingMinutes`	`REJECTED maxChargingMinutes`	**PASS**
BB-15	Charging limit lower boundary	FR-16	maxChargingMinutes=5; needs 8 min	`EXCLUDED DWELL_LIMIT`	`EXCLUDED DWELL_LIMIT`	**PASS**
BB-16	Charging time just above limit	FR-16	maxChargingMinutes=7.9; needs 8 min	`EXCLUDED DWELL_LIMIT`	`EXCLUDED DWELL_LIMIT`	**PASS**
BB-17	Charging time equals limit (boundary)	FR-16	maxChargingMinutes=8; needs 8 min	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-18	Charging limit upper boundary	FR-05	maxChargingMinutes=720	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-19	Charging limit invalid high	FR-05	maxChargingMinutes=720.1	`REJECTED maxChargingMinutes`	`REJECTED maxChargingMinutes`	**PASS**
BB-20	Manual target absent	FR-13, FR-14	manualTargetSoc=null	`targetSoc=53.33 taperWarning=false`	`targetSoc=53.33 taperWarning=false`	**PASS**
BB-21	Manual target invalid low	FR-14	manualTargetSoc=-0.1	`REJECTED manualTargetSoc`	`REJECTED manualTargetSoc`	**PASS**
BB-22	Manual target lower boundary	FR-14	manualTargetSoc=0	`EXCLUDED MANUAL_TARGET_TOO_LOW`	`EXCLUDED MANUAL_TARGET_TOO_LOW`	**PASS**
BB-23	Manual target just below automatic	FR-14	manualTargetSoc=53.3; automatic 53.33	`EXCLUDED MANUAL_TARGET_TOO_LOW`	`EXCLUDED MANUAL_TARGET_TOO_LOW`	**PASS**
BB-24	Manual target just above automatic	FR-14	manualTargetSoc=53.34; automatic 53.33	`targetSoc=53.34 taperWarning=false`	`targetSoc=53.34 taperWarning=false`	**PASS**
BB-25	Target at 80% (taper boundary)	FR-14	manualTargetSoc=80	`targetSoc=80.00 taperWarning=false`	`targetSoc=80.00 taperWarning=false`	**PASS**
BB-26	Target above 80% (taper boundary+)	FR-14	manualTargetSoc=80.1	`targetSoc=80.10 taperWarning=true`	`targetSoc=80.10 taperWarning=true`	**PASS**
BB-27	Manual target upper boundary	FR-14	manualTargetSoc=100	`targetSoc=100.00 taperWarning=true`	`targetSoc=100.00 taperWarning=true`	**PASS**
BB-28	Manual target invalid high	FR-14	manualTargetSoc=100.1	`REJECTED manualTargetSoc`	`REJECTED manualTargetSoc`	**PASS**
BB-29	Reserve invalid low	FR-04	reserveSoc=4.9	`REJECTED reserveSoc`	`REJECTED reserveSoc`	**PASS**

ID	Class / path	Trace	Test input	Expected output	Actual output	Result
BB-30	Reserve lower boundary	FR-13	reserveSoc=5; target 23 kWh, arrival 26 kWh	`EXCLUDED NO_CHARGING_REQUIRED`	`EXCLUDED NO_CHARGING_REQUIRED`	**PASS**
BB-31	Reserve upper boundary	FR-13	reserveSoc=40	`targetSoc=73.33 taperWarning=false`	`targetSoc=73.33 taperWarning=false`	**PASS**
BB-32	Reserve invalid high	FR-04	reserveSoc=40.1	`REJECTED reserveSoc`	`REJECTED reserveSoc`	**PASS**
BB-33	Capacity invalid low	FR-04	capacityKWh=9.9	`REJECTED capacityKWh`	`REJECTED capacityKWh`	**PASS**
BB-34	Capacity lower boundary	FR-13	capacityKWh=10; target 22 kWh	`EXCLUDED DESTINATION_UNREACHABLE`	`EXCLUDED DESTINATION_UNREACHABLE`	**PASS**
BB-35	Capacity upper boundary	FR-13	capacityKWh=250; arrival 121 kWh, target 70 kWh	`EXCLUDED NO_CHARGING_REQUIRED`	`EXCLUDED NO_CHARGING_REQUIRED`	**PASS**
BB-36	Capacity invalid high	FR-04	capacityKWh=250.1	`REJECTED capacityKWh`	`REJECTED capacityKWh`	**PASS**
BB-37	Target equals capacity (boundary)	FR-13	onward 240 km; needs exactly 60 kWh	`targetSoc=100.00 taperWarning=true`	`targetSoc=100.00 taperWarning=true`	**PASS**
BB-38	Target above capacity (boundary+)	FR-13	onward 241 km; needs 60.2 kWh	`EXCLUDED DESTINATION_UNREACHABLE`	`EXCLUDED DESTINATION_UNREACHABLE`	**PASS**
BB-39	Direct trip leaves exactly reserve	FR-13	direct 90 km; 12 kWh left, reserve 12 kWh	`noChargeNeeded=true`	`noChargeNeeded=true`	**PASS**
BB-40	Direct trip below reserve (boundary-)	FR-13	direct 90.5 km; 11.9 kWh left	`noChargeNeeded=false`	`noChargeNeeded=false`	**PASS**
BB-41	Compatible plug via alias	FR-10	vehicle "Combo 2"; station CCS2	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-42	No compatible plug	FR-10	vehicle CCS2; station CHAdeMO	`EXCLUDED NO_COMPATIBLE_CONNECTOR`	`EXCLUDED NO_COMPATIBLE_CONNECTOR`	**PASS**
BB-43	Window exactly fits opening hours	FR-11	openingHours=20:30-20:38	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-44	Opens one minute after arrival	FR-11	openingHours=20:31-21:00	`EXCLUDED CLOSED_DURING_CHARGING`	`EXCLUDED CLOSED_DURING_CHARGING`	**PASS**
BB-45	Closes before charging ends	FR-11	openingHours=20:00-20:37	`EXCLUDED CLOSED_DURING_CHARGING`	`EXCLUDED CLOSED_DURING_CHARGING`	**PASS**
BB-46	Overnight hours span midnight	FR-11	openingHours=20:00-06:00	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-47	Connector above vehicle limit	FR-15	connector 100 kW; vehicle DC 50 kW	`powerKw=50 min=8.00`	`powerKw=50 min=8.00`	**PASS**
BB-48	Connector below vehicle limit	FR-15	connector 40 kW; vehicle DC 50 kW	`powerKw=40 min=10.00`	`powerKw=40 min=10.00`	**PASS**
BB-49	Power equals minimum (boundary)	FR-15	minChargerKw=50; effective 50 kW	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
BB-50	Power below minimum (boundary-)	FR-15	minChargerKw=50.1; effective 50 kW	`EXCLUDED BELOW_MINIMUM_POWER`	`EXCLUDED BELOW_MINIMUM_POWER`	**PASS**
BB-51	Per-kWh price	FR-17	S$0.30/kWh	`cost=2.00`	`cost=2.00`	**PASS**
BB-52	Per-hour price	FR-17	S$0.80/h; 8 min	`cost=0.11`	`cost=0.11`	**PASS**
BB-53	Zero price (boundary)	FR-17	S$0/kWh	`cost=0.00`	`cost=0.00`	**PASS**
BB-54	Negative price (invalid)	FR-17	S$-0.01/kWh	`cost=null`	`cost=null`	**PASS**
BB-55	Unsupported price type	FR-17	priceType=free	`cost=null`	`cost=null`	**PASS**
BB-56	Strategy FASTEST	FR-18	Four-station ranking fixture	`A>C>B`	`A>C>B`	**PASS**
BB-57	Strategy CHEAPEST	FR-18	Four-station ranking fixture	`C>B>D`	`C>B>D`	**PASS**
BB-58	Strategy AVAILABILITY	FR-18	Four-station ranking fixture	`D>A>C`	`D>A>C`	**PASS**
BB-59	Strategy DETOUR	FR-18	Four-station ranking fixture	`B>A>C`	`B>A>C`	**PASS**
BB-60	Strategy BALANCED	FR-18	Four-station ranking fixture	`C>A>B`	`C>A>B`	**PASS**
BB-61	Strategy invalid	FR-05	strategy=RANDOM	`REJECTED strategy`	`REJECTED strategy`	**PASS**
BB-62	Three feasible stations (boundary)	FR-19	3 stations	`RECOMMENDED s1,s2,s3`	`RECOMMENDED s1,s2,s3`	**PASS**

ID	Class / path	Trace	Test input	Expected output	Actual output	Result
BB-63	Four feasible stations (boundary+)	FR-19	4 stations	`RECOMMENDED s1,s2,s3`	`RECOMMENDED s1,s2,s3`	**PASS**
BB-64	Departure without offset	FR-05	departureAt=2026-10-15T20:00:00	`singaporeAssumed=true`	`singaporeAssumed=true`	**PASS**
BB-65	Departure impossible date	FR-05	departureAt=2026-02-30T10:00:00+08:00	`REJECTED departureAt`	`REJECTED departureAt`	**PASS**
BB-66	Data age 9:59 (boundary-)	NFR-03	observedAt=11:50:01; now=12:00:00	`state=LIVE`	`state=LIVE`	**PASS**
BB-67	Data age 10:00 (boundary)	NFR-03	observedAt=11:50:00; now=12:00:00	`state=STALE`	`state=STALE`	**PASS**
BB-68	Fault expires after now	FR-23, FR-25	expiresAt=12:00:01; now=12:00:00	`EXCLUDED ACTIVE_FAULT_MARK`	`EXCLUDED ACTIVE_FAULT_MARK`	**PASS**
BB-69	Fault expires at now (boundary)	FR-23, FR-25	expiresAt=12:00:00; now=12:00:00	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**

3. Basis-path testing: failureForStation

This method applies every feasibility rule to one station. It is private, so each path is driven through calculatePlan with one station and observed from the result. **V(G) = 26 predicate nodes + 1 = 27.** A compound condition is one node because its parts decide a single branch. Ternaries used only to format text are not counted; the two that select calculation data (D12, D16) are.



Node	Predicate
D1	Route to or from the station is unusable
D2	Arrival battery energy < 0
D3	Detour > maximum detour
D4	No compatible connector

Node	Predicate
D14	Effective power < minimum
D15	Charging time > maximum
D16	Opening hours missing (assume all day)
D17	Window known and charging not covered

Node	Predicate
D5	Walking limit set, walking distance unusable
D6	Walking limit set, distance over limit
D7	No available compatible connector
D8	Automatic target > capacity
D9	Manual target below automatic target
D10	No charging required
D11	for each available connector
D12	Connector is AC (selects vehicle limit)
D13	Connector power invalid

Node	Predicate
D18	Price type is free
D19	Price type not per kWh or per hour
D20	Estimated cost is null
D21	Target SOC > 80%
D22	At least one candidate
D23	A connector hit closed hours
D24	A connector hit the charging limit
D25	A connector was below minimum power
D26	A connector had invalid power

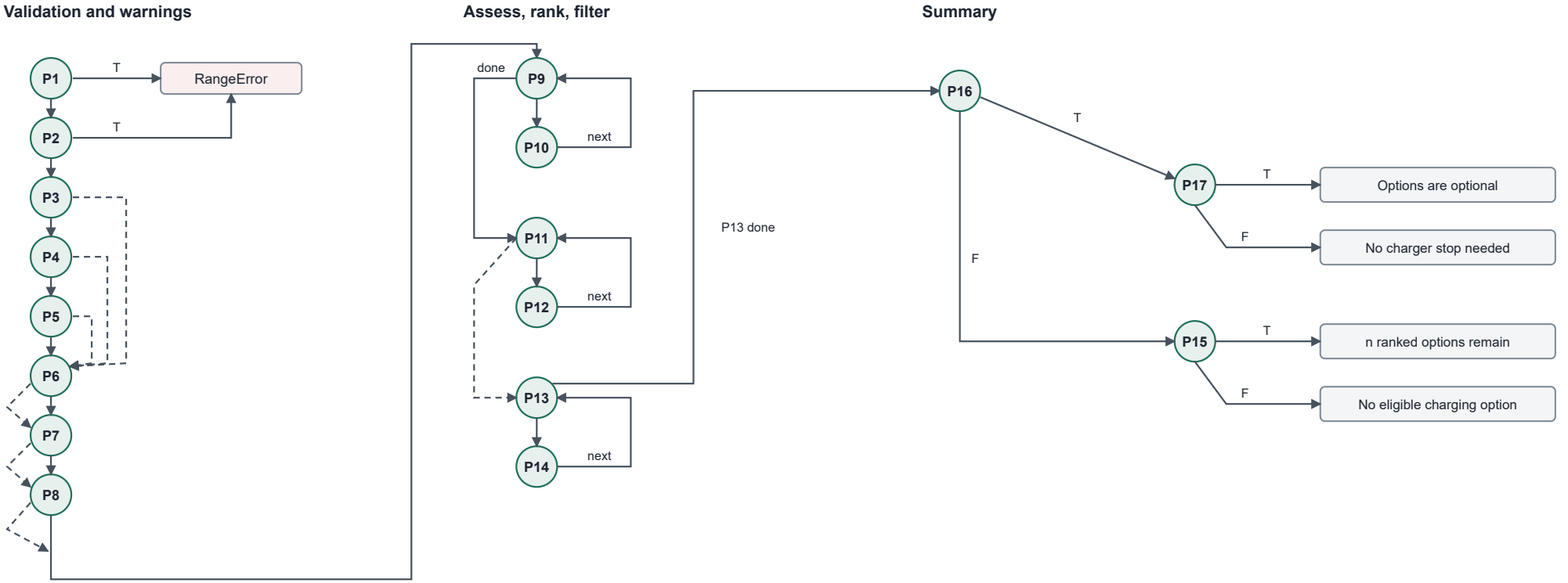
Starting from the baseline path, each case flips one decision. Solid arrows are control flow; dashed arrows rejoin after a warning or value choice.

ID	Class / path	Trace	Test input	Expected output	Actual output	Result
FS-01	Baseline: all gates pass, one DC connector, per-kWh price	Baseline	Baseline fixture	`RECOMMENDED s1`	`RECOMMENDED s1`	**PASS**
FS-02	D1 true: no usable route	D1	routes=[]	`EXCLUDED NO_ROUTE`	`EXCLUDED NO_ROUTE`	**PASS**
FS-03	D2 true: negative arrival energy	D2	to-station 200 km	`EXCLUDED UNREACHABLE`	`EXCLUDED UNREACHABLE`	**PASS**
FS-04	D3 true: detour over limit	D3	maxDetourMinutes=5; detour 10	`EXCLUDED DETOUR_LIMIT`	`EXCLUDED DETOUR_LIMIT`	**PASS**
FS-05	D4 true: no compatible connector	D4	CHAdeMO connector	`EXCLUDED NO_COMPATIBLE_CONNECTOR`	`EXCLUDED NO_COMPATIBLE_CONNECTOR`	**PASS**
FS-06	D5 true: walking distance unknown	D5	walkingMeters=undefined; limit 600	`EXCLUDED WALKING_UNVERIFIED`	`EXCLUDED WALKING_UNVERIFIED`	**PASS**
FS-07	D6 true: walking over limit	D6	walking 101 m; limit 100 m	`EXCLUDED WALKING_LIMIT`	`EXCLUDED WALKING_LIMIT`	**PASS**
FS-08	D7 true: no available connector	D7	connector OCCUPIED	`EXCLUDED NO_AVAILABLE_CONNECTOR`	`EXCLUDED NO_AVAILABLE_CONNECTOR`	**PASS**
FS-09	D8 true: target above capacity	D8	onward 241 km	`EXCLUDED DESTINATION_UNREACHABLE`	`EXCLUDED DESTINATION_UNREACHABLE`	**PASS**
FS-10	D9 true: manual target too low	D9	manualTargetSoc=50	`EXCLUDED MANUAL_TARGET_TOO_LOW`	`EXCLUDED MANUAL_TARGET_TOO_LOW`	**PASS**
FS-11	D10 true: no charging required	D10	onward 50 km; target 22 kWh, arrival 26 kWh	`EXCLUDED NO_CHARGING_REQUIRED`	`EXCLUDED NO_CHARGING_REQUIRED`	**PASS**
FS-12	D11 second iteration: two feasible connectors	D11	connectors 10 kW and 100 kW	`connector=fast`	`connector=fast`	**PASS**
FS-13	D12 true: AC connector uses AC limit	D12	AC 22 kW; vehicle AC 7 kW	`powerKw=7`	`powerKw=7`	**PASS**
FS-14	D13 true: invalid connector power	D13, D26	powerKw=0	`EXCLUDED INVALID_CONNECTOR_POWER`	`EXCLUDED INVALID_CONNECTOR_POWER`	**PASS**
FS-15	D14 true: below minimum power	D14, D25	minChargerKw=60	`EXCLUDED BELOW_MINIMUM_POWER`	`EXCLUDED BELOW_MINIMUM_POWER`	**PASS**
FS-16	D15 true: charging time over limit	D15, D24	maxChargingMinutes=7	`EXCLUDED DWELL_LIMIT`	`EXCLUDED DWELL_LIMIT`	**PASS**
FS-17	D16 true: opening hours missing	D16	openingHours=null	`assumedOpen=true`	`assumedOpen=true`	**PASS**
FS-18	D17 true: closed during charging	D17, D23	openingHours=09:00-17:00	`EXCLUDED CLOSED_DURING_CHARGING`	`EXCLUDED CLOSED_DURING_CHARGING`	**PASS**
FS-19	D18 true: free price type	D18	priceType=free	`freeWarning=true`	`freeWarning=true`	**PASS**
FS-20	D19 true: unsupported price type	D19	priceType=unknown	`unsupportedWarning=true`	`unsupportedWarning=true`	**PASS**
FS-21	D20 true: invalid unit price	D20	priceType=kWh; price=null	`invalidWarning=true`	`invalidWarning=true`	**PASS**
FS-22	D21 true: target above 80%	D21	manualTargetSoc=85	`taperWarning=true`	`taperWarning=true`	**PASS**

ID	Class / path	Trace	Test input	Expected output	Actual output	Result
FS-23	D22 true after a rejected connector	D22	connectors 0.5 kW and 100 kW	`connector=good`	`connector=good`	**PASS**
FS-24	D23 true while low power also seen	D23	09:00-17:00; connectors 100 kW and 0.5 kW	`EXCLUDED CLOSED_DURING_CHARGING`	`EXCLUDED CLOSED_DURING_CHARGING`	**PASS**
FS-25	D24 true while low power also seen	D24	minChargerKw=6; connectors 6 kW and 5 kW	`EXCLUDED DWELL_LIMIT`	`EXCLUDED DWELL_LIMIT`	**PASS**
FS-26	D25 true while invalid power also seen	D25	connectors 0.5 kW and 0 kW	`EXCLUDED BELOW_MINIMUM_POWER`	`EXCLUDED BELOW_MINIMUM_POWER`	**PASS**
FS-27	D26 false: loop rejected every connector, yet no flag set	D26	None exists	Infeasible. Every rejected connector sets one flag, and at least one connector is available here, so the false edge cannot execute.		**N/A**

4. Basis-path testing: calculatePlan

This method validates the request, classifies data freshness, ranks candidates, applies faults and writes the summary. **V(G) = 17 predicate nodes + 1 = 18.** Callbacks passed to filter and map are separate functions and are not counted.



Node	Predicate
P1	Direct route unusable
P2	Calculation or departure time invalid
P3	Snapshot stale
P4	Observation time unparseable
P5	Observation time in the future
P6	Retrieval time unparseable
P7	Calculation time has no offset
P8	Snapshot is Demo Fixture
P9	for each station

Node	Predicate
P10	Assessment has an exclusion
P11	for each ranked connector candidate
P12	Station has no best candidate yet
P13	for each faulted station
P14	Faulted station found in snapshot
P15	Options remain (charge needed)
P16	No charge needed
P17	Options remain (no charge needed)

ID	Class / path	Trace	Test input	Expected output	Actual output	Result
CP-01	Baseline: fresh LIVE, one feasible station	Baseline	Baseline fixture	`state=LIVE recs=1 summary=1 ranked charger option remain after feasibility and fault checks.`	`state=LIVE recs=1 summary=1 ranked charger option remain after feasibility and fault checks.`	**PASS**
CP-02	P1 true: unusable direct route	P1	directRoute.distanceKm=-1	`ERROR RangeError`	`ERROR RangeError`	**PASS**
CP-03	P2 true: invalid calculation time	P2	now=not-a-time	`ERROR RangeError`	`ERROR RangeError`	**PASS**
CP-04	P3 true: stale snapshot	P3	observedAt=11:40	`state=STALE warning=true`	`state=STALE warning=true`	**PASS**
CP-05	P4 true: unparseable observation time	P4	observedAt=not-a-time	`warning=true`	`warning=true`	**PASS**
CP-06	P5 true: observation in the future	P5	observedAt=12:05	`warning=true`	`warning=true`	**PASS**
CP-07	P6 true: unparseable retrieval time	P6	retrievedAt=not-a-time	`warning=true`	`warning=true`	**PASS**
CP-08	P7 true: calculation time without offset	P7	now=2026-10-15T12:00:00	`warning=true`	`warning=true`	**PASS**
CP-09	P8 true: Demo Fixture snapshot	P8	state=DEMO_FIXTURE	`state=DEMO_FIXTURE warning=true`	`state=DEMO_FIXTURE warning=true`	**PASS**
CP-10	P9 zero iterations: empty station list	P9	stations=[]	`NO_OPTIONS`	`NO_OPTIONS`	**PASS**
CP-11	P11 zero iterations: every station excluded	P11, P15	one station, connector OCCUPIED	`summary=No eligible charging option remains; review the exclusions or change the plan inputs.`	`summary=No eligible charging option remains; review the exclusions or change the plan inputs.`	**PASS**
CP-12	P10 true with a ranked candidate	P10	one excluded and one feasible station	`recs=s2 excluded=NO_AVAILABLE_CONNECTOR`	`recs=s2 excluded=NO_AVAILABLE_CONNECTOR`	**PASS**
CP-13	P12 false: second connector of a station	P12	one station, two feasible connectors	`recs=1 connector=c2`	`recs=1 connector=c2`	**PASS**
CP-15	P13 entered: fault removes one of two stations	P13	active fault on s1; s1 and s2 feasible	`RECOMMENDED s2`	`RECOMMENDED s2`	**PASS**
CP-16	P15 false: fault removes the only station	P15	active fault on s1; only station	`EXCLUDED ACTIVE_FAULT_MARK`	`EXCLUDED ACTIVE_FAULT_MARK`	**PASS**
CP-17	P16 true, P17 true: enough charge, options remain	P16, P17	direct 50 km	`summary=Current charge is sufficient to reach the destination with reserve; charger options are optional.`	`summary=Current charge is sufficient to reach the destination with reserve; charger options are optional.`	**PASS**
CP-18	P16 true, P17 false: enough charge, no options	P17	direct 50 km; stations=[]	`summary=Current charge is sufficient to reach the destination with reserve; no charger stop is needed.`	`summary=Current charge is sufficient to reach the destination with reserve; no charger stop is needed.`	**PASS**
CP-14	P14 false: faulted station not in snapshot	P14	None exists	Infeasible. Faulted stations are filtered to ranked stations, and every ranked station comes from the snapshot.		**N/A**